SESSION 5

FINANCIAL PLANNING & ANALYSIS – 1

SCALER SCHOOL OF TECHNOLOGY

Learning Outcomes

Session 5: Introduction to Management Accounting

By the end of the session, learners will be able to:

- **Explain** the role and importance of Management Accounting in decision-making.
- **Classify** different cost concepts and cost behaviours.
- **Apply** cost concepts to basic business and engineering scenarios.
- **Calculate** and interpret basic cost information.
- **Analyse** cost behaviour to support cost control and decision-making.

Management Accounting

Introduction:

Financial Accounting deals with financial information & is based on **GAAP (General Accepted Accounting Principles).**

Managerial Accounting is the process of **identifying, measuring, analysing, and communicating financial and non-financial information** to managers for decision-making. It supports the following activities:

- ❑ **Planning** – Estimating the costs, revenues, profits & resources **(Budgeting)**
- ❑ **Control** – Monitoring whether plans are being achieved **(Actual performance Vs Budgeted Performance & Variance Analysis)**
- ❑ **Decision-making** – choosing the best available alternative
  - ✔ Make or buy
  - ✔ Outsource or produce in-house
  - ✔ Product selection
  - ✔ Pricing **(competitive pricing, penetration pricing, cost-plus pricing, full cost pricing, marginal cost pricing, price skimming)**

A key component of managerial accounting is **cost accounting.**

Management Accounting

Cost Concepts:

What is Cost: It is a **sacrifice of resources,** an amount paid, to achieve a specific objective such as **acquiring an asset, manufacturing a product or providing a service.**

However,

$$\text{Cost} \neq \text{Expense}$$

$$\text{Expense} = \text{Cost}$$

Expense is a cost against revenue in a particular accounting period

Actual cost is a cost that has already occurred.

Budgeted cost is an estimated cost.

A cost object measures the resources consumed by any part of a business. It could be a product, service, project, customer, activity, department, or distribution channel.

Costing is the process of measuring the cost of cost objects.

Management Accounting

Cost Classification: It means **arranging or grouping costs** into logical categories based on a particular characteristic or purpose.

A **single cost** can be **classified in different ways** depending on the **purpose** of the analysis.

Example: The cost of steel used to manufacture a car can be classified as:

- **By nature:** Material cost
- **By function:** Production cost
- **By behaviour:** Variable cost
- **By traceability:** Direct cost
- **By relevance:** Relevant to a production decision

Cost Classification helps management to determine the cost of a good or service.

Management needs to know how much it costs to produce one unit.

For example, if total production costs are ₹50 lakh and 10,000 units are produced, then the **cost per unit** is,

$$\text{Cost per unit} = \frac{₹50,00,000}{10,000} = ₹500$$

Management Accounting

Methods of Cost Classification:

Classification by Nature/Element: Costs are classified according to what the cost represents.

- **Material Cost:** Cost of materials used in production.
  - ✔ Steel used in cars
  - ✔ Wood used in furniture
  - ✔ Fabric used in garments
  - ✔ Electronic components used in smartphones

Material costs can further be classified as:

- **Direct Material:** It can be directly identified with a product. **(steel body used in cars)**
- **Indirect Material:** It cannot be economically traced to a particular product. **(Lubricants used in the car factory)**

- **Labour Cost:** Cost of labour used in production or employee services:
  - ✔ Salary of assembly-line workers
  - ✔ Wages of machine operators
  - ✔ Salary of production supervisors

Labour costs can further be classified as:

- **Direct Labour:** Labour directly involved in producing the product. **(Assembly-line worker)**
- **Indirect Labour:** Labour supporting production but not directly traceable to individual units. **(Factory security guard)**

- **Other Expense:** Costs other than labour or material.

Management Accounting

Classification by Function: Costs are classified according to the **business function in which they are incurred.**

- **Production or Manufacturing Cost:** Cost incurred in **manufacturing** a product or providing a service.
  - ✔ Raw materials
  - ✔ Production workers' wages
  - ✔ Factory electricity
  - ✔ Factory depreciation
  - ✔ Factory rent

- **Administration Cost:** Cost associated with **general administration & management.**
  - ✔ Senior management salary
  - ✔ Office insurance
  - ✔ Depreciation of office equipment
  - ✔ Head office rent
  - ✔ Audit charges
  - ✔ Office supplies
  - ✔ Legal fees

Management Accounting

Classification by Function: Costs are classified according to the **business function in which they are incurred.**

- **Selling Cost:** Cost incurred to **generate sales**.
  - ✔ Sales commission
  - ✔ Advertising & other promotional expenses

- **Distribution Cost:** Costs associated with **delivering products to customers.**
  - ✔ Warehousing
  - ✔ Transportation
  - ✔ Delivery charges

Functional Classification:

$$\text{Total Cost} = \text{Production Cost} + \text{Administration Cost} + \text{Selling Cost} + \text{Distribution Cost}$$

Management Accounting

Classification by Traceability: Costs can be **directly traced to a specific product, service, department or cost object.**

These costs are of two major categories:

- **Direct Cost:** These costs can be **traced to a specific cost object.**
  - ✔ Steel used for manufacturing a particular car
  - ✔ Timber used for a particular table
  - ✔ Wages of a worker assembling a specific product

- **Indirect Cost:** These costs **cannot be traced to a specific cost object.**
  - ✔ Factory rent
  - ✔ Factory electricity
  - ✔ Factory insurance
  - ✔ Supervisor's salary

Indirect costs are commonly referred to as **overheads.**

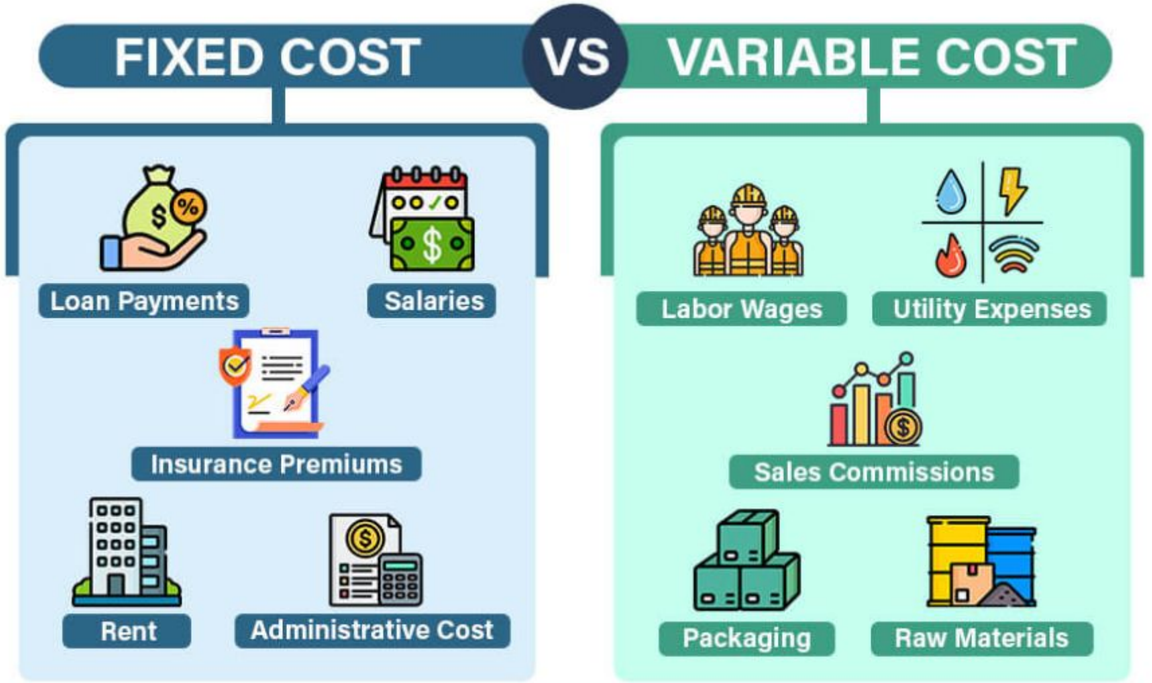
Management Accounting

Classification by Behaviour: Cost behaviour refers to **how a cost changes when the level of activity changes**.

These costs are of three major categories:

- **Fixed Cost:** These **costs remain constant or fixed** in total within a relevant range**, regardless of changes in activity.**
  - ✔ Factory rent
  - ✔ Manager's salary
  - ✔ Insurance

The **fixed cost per unit decreases** as production increases.



Management Accounting

Fixed Cost: They remain fixed in total, but decrease per unit

Example: Suppose an automobile company pays ₹10,00,000 monthly factory rent. The rent does not change with the number of cars produced, as long as production remains within the factory's existing capacity.

Cars produced	Total Fixed Cost	Fixed Cost per Car
1,000	₹10,00,000	₹1,000
2,000	₹10,00,000	₹500
5,000	₹10,00,000	₹200
10,000	₹10,00,000	₹100

Total fixed cost remains constant: **Fixed Cost = ₹10,00,000**

But fixed cost **per unit decreases** as production increases:

$$\text{Fixed Cost per Unit} = \frac{\text{Total Fixed Cost}}{\text{Number of Units}}$$

$$\frac{₹10,00,000}{1,000} = ₹1,000$$



$$\frac{₹10,00,000}{10,000} = ₹100$$

Management Accounting

Classification by Behaviour: Cost behaviour refers to **how a cost changes when the level of activity changes**.

- **Variable Cost:** These costs **change in total in proportion to changes in activity.**
  - ✔ Direct materials
  - ✔ Direct labour paid per unit
  - ✔ Sales commission
  - ✔ Delivery cost per unit
  - ✔ Power consumed in production

The total variable cost changes, but **variable cost per unit** remains **constant.**

Management Accounting

Variable Cost: They remain variable in total, but constant per unit

Example: Suppose a company uses raw material costing ₹20 for each bottle produced.

Therefore, **Variable Cost per Unit = ₹20**

Bottles Produced	Variable Cost per Bottle	Total Variable Cost
1,000	₹20	₹20,000
2,000	₹20	₹40,000
5,000	₹20	₹1,00,000
10,000	₹20	₹2,00,000

The **total variable cost increases** as production increases: From ₹20,000 for 1,000 bottles to ₹40,000 for 2,000 bottles and so on…

But the **variable cost per unit remains ₹20.**

$$\text{Variable Cost} = \text{Variable Cost per Unit} \times \text{Output}$$

For 5,000 bottles: $₹20 \times 5,000 = ₹1,00,000$

For 10,000 bottles: $₹20 \times 10,000 = ₹2,00,000$

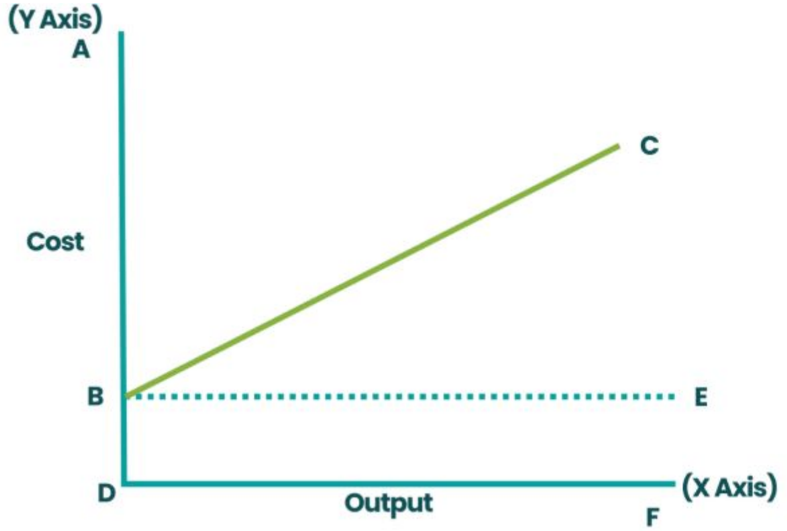
Management Accounting

Classification by Behaviour: Cost behaviour refers to **how a cost changes when the level of activity changes**.

- **Semi–variable cost:** A **semi-variable cost** is a cost that contains **both a fixed component and a variable component.**
  - ✔ Factory rent
  - ✔ Factory manager's salary
  - ✔ Factory security guard
  - ✔ Factory depreciation

The **fixed component** is incurred even when **activity is zero,** while the variable component increases as the level of activity increases.

Semi-variable Cost = Fixed Cost + Variable Cost



Management Accounting

Semi–Variable Cost:

Example: Suppose a manufacturing plant has an electricity bill consisting of:

- **Fixed** monthly connection/demand charge = **₹50,000**
- **Variable** electricity cost = **₹8 per machine-hour**

Machine Hours	Fixed Component	Variable Component	Total Electricity Cost	Cost per Machine Hour
1,000	₹50,000	₹8,000	₹58,000	₹58.00
2,000	₹50,000	₹16,000	₹66,000	₹33.00
5,000	₹50,000	₹40,000	₹90,000	₹18.00
10,000	₹50,000	₹80,000	₹1,30,000	₹13.00

At 5,000 machine-hours:

$$\text{Total Cost} = ₹50,000 + (₹8 \times 5,000)$$

$$= \boxed{₹90,000}$$

Management Accounting

Classification by Relevance: These are costs based on whether they are **relevant to a particular decision.**

- **Relevant Cost:** A **future cost** that **changes between alternatives.**
- **Irrelevant Cost:** A cost that **does not change between alternatives.**
- **Sunk Cost:** A cost that has already been incurred and **cannot be recovered.**

Example: A technology company spends **₹2 crore** developing a new software platform.

After development, management discovers that competitors have launched a superior product.

The company now has two options:

- **Option A:** Launch the software
- **Option B**: Abandon the project

The ₹2 crore has already been spent.

Therefore, ₹2 crore = Sunk Cost

It should not determine whether the company launches the product.

Management Accounting

Classification by Relevance: Costs are classified by whether they are **relevant to a particular decision.**

- **Opportunity Cost:** The **benefit sacrificed** by **choosing one alternative over another.** In other words, it is the **cost of the best alternative forgone.**

Opportunity Cost = Benefit sacrificed from the next best alternative

Example: A student has the following 2 options:

- Accept a job with an annual salary of ₹16 lakh, or
- Pursue a full-time MBA.

✔ If the **student chooses the MBA**, **the opportunity cost is the ₹16 lakh salary forgone.**
✔ If the **student chooses a job,** the opportunity cost is **the benefit of doing an MBA,** which would have provided **better managerial jobs in the future.**

Management Accounting

Cost Unit: A **cost unit** is a unit of product or service in relation to which **costs can be ascertained, accumulated and expressed.**

Example:

- **Automobile manufacturer**: per car
- **Hotel:** per room-night
- **Airline:** per passenger-kilometre
- **Electricity company:** per kilowatt-hour
- **Hospital:** per patient-day
- **Transport company:** per tonne-kilometre

Management Accounting

Absorption of Overheads: A **business** needs to determine the **cost per unit of goods or services produced** for several important reasons.

- ✔ Value **inventory** (stock)
- ✔ Determine **selling prices**
- ✔ Measure **profitability**
- ✔ Compare **product profitability**
- ✔ Control and monitor **costs**
- ✔ Make **production and outsourcing** decisions

For example, if a company manufactures a product at a **total cost of ₹80 per unit,** management needs to know this cost before deciding whether a **selling price of ₹100 or ₹120** is appropriate.

Management Accounting

Components of Product Cost: For a manufacturing business, the cost of producing a product generally consists of:

- **Direct Materials:** Materials that can be directly traced to a product.
  Example: Aluminium used to manufacture a car.

- **Direct Labour:** Labour directly involved in manufacturing the product.
  Example: Wages of an employee assembling the machine.

- **Overheads:** Other **costs** incurred in the production process that **cannot be directly traced** to the product or service.
  Example:

✔ Factory rent
✔ Factory electricity
✔ Factory insurance
✔ Production supervisor's salary
✔ Factory maintenance
✔ Factory security

Total Production Cost = Direct Material + Direct Labour + Production Overheads

Management Accounting

Example: An engineering company manufactures electric motors.

Each motor requires:

- **Direct material =** ₹4,000
- **Direct labour =** ₹2,000

Therefore, the **Direct cost per motor = ₹6,000**

However, the factory also incurs:

✔ Factory rent
✔ Electricity
✔ Machine depreciation
✔ Supervisor's salary
✔ Factory maintenance
✔ Security

These are production overheads.

The question is: ***How much of these overheads should be included in the cost of each motor?***

Management Accounting

Overhead Absorption: It is the process of **charging a share of total production overheads to individual units** of production or other cost units **using an appropriate absorption rate.**

In simple words, ***overhead absorption means spreading the cost of overheads across the products or services that benefit from those overheads.***

Example:

- ✔ Total factory overhead = $100,000
- ✔ **Units produced =** 20,000

If units are considered an appropriate basis:

$$\text{Overhead per unit} = \frac{\$100,000}{20,000} = \$5$$

Therefore, **$5 of factory overhead is absorbed into each unit**

The **Overhead Absorption Rate (OAR)** is calculated as follows:

$$\text{OAR} = \frac{\text{Budgeted Overhead}}{\text{Budgeted Activity}}$$

Management Accounting

Marginal Cost (MC): Marginal cost is the **additional cost incurred** as a result of **producing one additional unit** of output.

✔ In practical Management Accounting, **marginal cost** is generally associated with the **variable cost of producing an additional unit,** provided that fixed costs do not change.

$$\text{Marginal Cost} = \frac{\text{Change in Total Cost}}{\text{Change in Output}}$$

Example:
An engineering company produces **10,000 components** at a total cost of **₹50,00,000.**

If it increases production to **10,100 components, the** total cost becomes **₹50,60,000**

Increase in output: 10,100 – 10,000 = **100 units**

Increase in cost: ₹50,60,000 - ₹50,00,000 = **₹60,000**

Therefore,

$$MC = \frac{₹60,000}{100} = \boxed{₹600 \text{ per additional component}}$$

So, producing the additional 100 components costs **₹600 per component,** on average.

Management Accounting

Marginal Costing (MC): Marginal costing is a **costing technique** in which **variable costs are charged to products** and **fixed costs are treated as period costs.**

$$\text{Marginal Costing} \neq \text{Marginal Cost}$$

✔ The central idea in **Marginal Costing** is to calculate **Contribution.**

$$Contribution = Sales - Variable\ Cost$$

At the unit level:

$$Contribution\ per\ Unit = Selling\ Price - Variable\ Cost\ per\ Unit$$

Since marginal cost is normally equal to variable cost per unit:

$$Contribution\ per\ Unit = Selling\ Price - Marginal\ Cost$$

Management Accounting

When there is no capacity constraint:

Example: An airline has one empty seat on a flight
The **normal ticket price** is **₹8,000.**
Most flight costs, such as aircraft lease, crew salaries, airport charges, etc., are already committed.

The additional cost of carrying one more passenger is estimated at:

- **Meal: ₹500**
- **Passenger service: ₹200**
- **Processing cost: ₹100**
- Other **variable cost: ₹200**

Marginal Cost = ₹1,000

A customer is willing to pay **₹3,000** for the **last-minute seat.**

Additional contribution: **₹3,000 – ₹1,000 = ₹2,000**

Decision: **Sell the seat for ₹3,000,** assuming it would otherwise remain empty.

Management Accounting

When there is a capacity constraint: A **capacity constraint** occurs when a business has a **limited resource that restricts the amount of goods or services it can provide.**

- ✔ In other words, **a capacity constraint means that the business has more potential demand than it has the capacity to serve.**
- ✔ The scarce resource that creates this restriction is called a **limiting factor** or **bottleneck.**

$$\text{Demand or Required Capacity} > \text{Available Capacity}$$

Management Accounting

When there is a capacity constraint:

Example:

An airline is operating a **Bengaluru – Mumbai** flight. The following details are available:

- The aircraft has **180 seats.**
- **175 seats** have already been sold at an average fare of **₹8,000**
- Only **5 seats** remain available.
- A **travel company** approaches the airline and **offers to purchase 10 seats at ₹3,000 per passenger.**
- The airline cannot sell all 10 seats because **only 5 seats are available.**
- The additional cost of carrying one passenger is estimated as follows:
  - ✔ **Meal:** ₹500
  - ✔ **Passenger service:** ₹200
  - ✔ **Processing charges:** ₹100
  - ✔ **Other variable cost:** ₹200

The airline expects that if it does not sell the remaining seats to the travel company, it can sell them to individual passengers at **₹7,000 per seat.**

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Management Accounting

When there is a capacity constraint:

Required:

1. Calculate the **marginal cost per passenger.**
2. Calculate the **contribution per passenger** if the airline accepts the **travel company's offer of ₹3,000.**
3. Calculate the **contribution per passenger** from an **individual passenger paying ₹7,000.**
4. Calculate the **opportunity cost of allocating a seat to the travel company.**
5. Determine the **minimum price the airline should accept for the remaining seats** under the capacity constraint.
6. Should the **airline accept** the travel company's offer **of ₹3,000 per seat?**

Solution:

- Airline total seating capacity = **180**
- Seats already sold = **175**
- Remaining (spare) seats = **180 – 175 = 5**

The travel company wants 10 seats, but only 5 are available.

Therefore, **capacity constraints = 5 available seats**

Management Accounting

When there is a capacity constraint:

1. **Marginal cost per passenger:**
   Additional costs per passenger: ₹ 500 + ₹ 200 + ₹ 100 + ₹ 200 **= ₹1,000**

2. **Contribution per passenger** if the airline accepts the **travel company's offer of ₹3,000**
   Travel company's proposed price = ₹3,000
   Marginal cost = ₹1,000

   Therefore, Contribution = Sales – Variable Cost
   = ₹3,000 – ₹1,000 **= ₹2,000 per passenger**

If the airline sells all 5 available seats to the travel company, then:
The total contribution = 5 x ₹2,000 **= ₹10,000**

3. **Contribution per passenger** from an **individual passenger paying ₹7,000**

   Expected fare from an individual passenger = ₹7,000
   Marginal cost = ₹1,000

   Therefore, Contribution = Sales – Variable Cost
   = ₹6,000 – ₹1,000 **= ₹7,000 per passenger**

Management Accounting

When there is a capacity constraint:

4. The opportunity cost of allocating a seat to the travel company

If the airline **sells a seat** to the travel company for **₹3,000,** it gives up **the opportunity to sell** that seat to an individual passenger for **₹7,000.**

Therefore, the contribution forgone = ₹7,000 - ₹1,000 = ₹6,000 (Opportunity cost per seat)

5. The Minimum Acceptable Price

When there is a capacity constraint, **Minimum Acceptable Price = Marginal Cost + Opportunity Cost of foregone opportunity**

Therefore, the minimum acceptable price = ₹1,000 + ₹6,000 = ₹7,000

6. Should the airline accept airline accept the travel company's offer of ₹3,000 per seat?

The airline will reject the offer from the travel company, as accepting the travel company's offer would result in ₹20,000 [5 x (6000 – 2000)] less contribution than selling the five seats to individual passengers.

Thank You!